

Luis Alberto Núñez de Villavicencio y Martínez. Updated May 1th, 2026

■ Personal Data

- *Place and Date of Birth*: La Habana, Cuba, 14 Feb 1957; *Nationality*: Venezuelan and Colombian.
- *ORCID*: <https://orcid.org/0000-0003-4575-5899>
- *Scholar*: https://scholar.google.co.ve/citations?user=2Q5_QxkAAAAJ&hl=en

■ Fields of Interest:

General Relativity, Relativistic Astrophysics, Computational Physics and Information Technology, Astro-particle physics, Member of the Latin American Giant Observatory (LAGO) and Pierre Auger Observatory

■ Education:

(1983 - 1988) *Graduate* Univ. Central de Venezuela, Caracas-Venezuela *Doctor en Ciencias*;
(1973 - 1979) *Undergraduate* Univ. Simón Bolívar, Caracas-Venezuela *Licenciado en Física*.

■ Quantitative indicators of academic trajectory (updated May 2026)

Thesis Supervisor:

PhD 5: four in Physics and one in Electrical Engineering;
Master 24

2 x TFM (Spain); 2 x Geophysics; 2 x System Engineering, 2 in Applied Mathematics; 16 in Physics.

Total publications 1983-2026 (Scopus): 304; journal papers 131, conference papers 158

Total citations 1983-2026 (Google Scholar): 11666; h-index = 52; i100= 24; i10-index = 153

■ Academic Positions

(2010 - Present) **Full Professor**, Escuela de Física, Univ. Industrial de Santander, Colombia,
(1979 - 2009) **Emeritus Professor**. Departamento de Física, Univ. de Los Andes, Venezuela.

■ Participation in international projects:

(2025-Date) Co-PI [E-Latinoamerican HuB for academic grOwiNG cOmmunities in physics, EL-BONGO physics](#), Erasmus+ CBHE Project.

(2024-Date) PI [LAGO-INDICA. Infraestructura digital de ciencia abierta \(524RT0159\) Network](#), Ciencia y Tecnología para el Desarrollo (CyTED).

(2023-Date) PI Latin America Gigant Observatory (LAGO) [Latin America Gigant Observatory \(LAGO\)](#)

(2020-2024) Co-PI [Latin-American alliance for Capacity buildiNG in Advanced physics, LA-CoNGA physics](#); Erasmus+ CBHE Project.

(2018-2019) Co-Pi [Muon tomography of volcanoes in South America](#). UK Innovation and Research.

(2012-2014) Responsible for the infrastructure Workpackage [Europe Latin America Collaborative e-Infrastructure for Research Activities \(ELCIRA\)](#); 7framework Programme, European Commission.

(2012-2015) Board of Partners [Coordination and Harmonisation of Advanced e-Infrastructures for Research and Education Data Sharing \(CHAIN-REDS\)](#); 7framework Programme, European Commission.

(2010-2012) Board of Partners [Co-ordination and Harmonisation of Advanced e-Infrastructures CHAIN](#); 7framework Programme, European Commission.

(2010-2012) Technical coordinator [Grid Initiatives for E-Science Virtual Communities in Europe and Latin America \(GISELA\)](#); 7framework Programme, European Commission.

(2008-2010) Board of Partners [E-Science Grid Facility for Europe and Latin America EELA2](#) 7framework Programme, European Commission.

(2006-2007) Board of Partners [E-Infrastructure shared between Europe and Latin America EELA](#) 7framework Programme, European Commission.

(2005-2006) Country representative [High Energy Physics Latin-American-European Network \(HELEN\)](#) Alpha Program. European Commission.

- **Administrative Positions:**

(2011- 2014) **Academic Manager** [RedCLARA](#), Corporación Latinoamericana de Redes Avanzadas
 (1997 - 2009) **Director of the National Center for Scientific Computing CeCalCULA** Tech Park of Mérida Mérida-Venezuela.
 (1995 - 2009) **Director of Academic Computing**, Univ. de los Andes Mérida Venezuela.
 (2003 - 2006) **Member of the Board of Directors of National Center for Information Technology**. Centro Nacional de Tecnologías de Información (CNTI) Venezuela

- **Distintions:**

(2026) Eloy Valenzuela Award for the best paper (*Design and construction of MuTe: a hybrid muon telescope to study Colombian volcanoes*) of the Science Faculty, Univ. Industrial de Santander, Colombia.
 (2024) Mateo Valero International Award in High Performance Computing, Sistema de Computación Avanzada para Latinoamérica y el Caribe (SCALAC).
 (2024) International collaborator Gravitación y Cosmología Relativista research group, Instituto Universitario de Física Fundamental y Matemáticas, Univ. de Salamanca, Spain.
 (2020) Eloy Valenzuela Award as a consolidated researcher, Univ. Industrial de Santander, Colombia.
 (2014) Best teacher of the Science Faculty, Univ. Industrial de Santander, Colombia

- **Paper in journals 2020-2026**

These papers, with fewer than 10 authors, have to be added to the Pierre Auger Collaboration, which includes 78 publications.

2026

⇒ Sarmiento-Cano, C., Betancour, J., Núñez-Selin, A., Miranda-Leuro, L., Sidelnik, I., Asorey, H., Núñez, L.A. [Water Cherenkov detectors in precision agriculture: A novel approach for high-resolution soil moisture monitoring](#), *Nuclear Instruments and Methods in Physics Research Section A*, **1090**, 171620.
 ⇒ Villamizar-Mantilla, D.A., Núñez, L.A., Stashenko, E.E., Vinardell, M.P., Fuentes, J.L. [Erythema Protection Efficacy of Plant-Derivative Compounds](#) *Life*, **16**(1), 176.
 ⇒ Núñez Selin, A., Sidelnik, I., Sarmiento-Cano, C., Asorey, H., Núñez, L.A. [Two stage gamma neutron source classification in water Cherenkov detectors: Energy threshold screening and machine learning pulse analysis](#), *Nuclear Instruments and Methods in Physics Research Section A*, **1088**, 171517.

2025

⇒ Martínez-Rivero, R. A., Sarmiento-Cano, C., Catillo-Morales, D., Perea-Pérez, J., Baldovino-Medrano, V., Sanabria-Gómez, J. D., and Núñez, L. A. (2025). [Muon Imaging of Hydrotreatment Reactors](#). *Journal of Applied Physics*, **138** (24). and also [arXiv preprint arXiv:2504.15103](#).
 ⇒ Ospino, J., Hernández-Pastora, J. L., Araujo-Salcedo, A. V., and Núñez, L. A. (2024). [A Methodological Framework for Solving Einstein's Equations in Axially Symmetric Spacetimes](#). *The European Physical Journal Plus*, **140**(5), 414 and [arXiv preprint arXiv:2412.15480](#) and
 ⇒ Ospino, J., Suárez-Urango, D., Becerra, L. M., Hernández, H., and Núñez, L. A. (2024). [Relativistic non-pascalian fluid as a density contribution](#) *The European Physical Journal C* **85**, 412 (2025) and [arXiv preprint arXiv:2410.18231](#).

2024

⇒ Peña-Rodríguez, J., Jaimes-Teherán, J., Dlaikan-Castillo, K., Núñez, L. A. (2024). [MUYS: An end-to-end muography simulation toolbox](#). *Geophysical Journal International*, **327** (1) 540-556. also [arXiv preprint arXiv:2303.02627](#).

2023

⇒ Becerra, L. M., Sarmiento-Cano, C., Martínez-Méndez, A., Dominguez, Y., and Núñez, L. A. (2023). [High-Performance Computing for Astrophysical Simulations and Astroparticle Observations](#). In Latin American High Performance Computing Conference (pp. 184-196). Cham: Springer Nature Switzerland.
 ⇒ Suárez-Urango, D., Becerra, L. M., Ospino, J., Núñez, L. A. (2023). [The physical acceptability conditions and the strategies to obtain anisotropic compact objects](#). *The European Physical Journal C*, **83**(11), 1018.

⇒ Peña-Rodríguez, J., Sánchez-Villafrades, J., Asorey, H., Núñez, L. A. (2023). [Characterization and On-Field Performance of the MuTe Silicon Photomultipliers](#). *Instruments*, **7**(1), 7.

⇒ Sierra-Porta, D., Solano-Correa, Y.T., Tarazona-Alvarado, M., Núñez, L.A. (2023) [Linking PM10 and PM2.5 Pollution Concentration Through Tree Coverage in Urban Areas](#). *CLEAN: Soil, Air, Water* **51**(5), 2200222

2022

⇒ Sarmiento-Cano, C., Suárez-Durán, M., Calderón-Ardila, R., Vásquez-Ramírez, A., Jaimes-Motta, A., Dasso, S., Sidelnik, I., Núñez, L.A. and Asorey H.(2022). [The ARTI framework: cosmic rays atmospheric background simulations](#). *The European Physical Journal C*, **82**(11), 1019.

⇒ Ospino, J., Hernandez-Pastora, J. L., and Núñez, L. A. [All analytic solutions for geodesic motion in axially symmetric space-times](#). *The European Physical Journal C* **82** 591 (2022).

⇒ D. Suárez-Urango, J. Ospino, H. Hernández, and L.A. Núñez. [Acceptability conditions and relativistic anisotropic generalized polytropes](#). *The European Physical Journal C* **82**, 176 (2022).

⇒ J. Grisales-Casadiegos, C. Sarmiento-Cano, and L.A. Núñez. [Impact of global data assimilation system atmospheric models on astroparticle showers](#). *Canadian Journal of Physics* **100**(3), 1-6. (2022).

⇒ Peña-Rodríguez, J., Vesga-Ramírez, A., Vásquez-Ramírez, A., Suárez-Durán, M., de León-Barrios, R., Sierra-Porta, D., Calderón-Ardila, R., Pisco-Guavabe, J., Asorey, H. and Núñez, L. A. (2022). [Muography in Colombia: Simulation Framework, Instrumentation, and Data Analysis](#). *Journal of Advanced Instrumentation in Science*, JAIS-271.

2021

⇒ J. Peña-Rodríguez, S. Hernández-Barajas, Y. León-Carreño, and L.A. Núñez. [Modeling and simulation of the r5912 photomultiplier for the lago project](#). *IEEE Sensors Journal*, **21**(18):20184–20191, (2021).

⇒ J.L. Fuentes, D.A. Villamizar-Mantilla, S.J. Flores-González, L.A. Núñez, and E.E. Stashenko. [Plants growing in colombia as sources of active ingredients for sunscreens](#). *International Journal of Radiation Biology*, **97**(12):1705-1715, (2021).

⇒ H. Hernández, D. Suárez-Urango, and L.A. Núñez. [Acceptability conditions and relativistic barotropic equations of state](#). *Eur. Phys. J. C*, **81**(241), 2021.

⇒ A. Vesga-Ramírez, J.D. Sanabria-Gómez, D. Sierra-Porta, L. Arana-Salinas, H. Asorey, V.A. Kudryavtsev, R. Calderón-Ardila, and L.A. Núñez. [Simulated annealing for volcano muography](#). *Journal of South American Earth Sciences*, **109**:103248, (2021).

2020

⇒ A. Martínez-Méndez and L.A. Núñez. [Academia, datos y reproducibilidad de la ciencia](#). *Revista UIS Ingenierías*, **19** (4):315–324, (2020).

⇒ A. Vesga-Ramírez, D. Sierra-Porta, J. Peña-Rodríguez, J.D. Sanabria-Gómez, M. Valencia-Otero, C. Sarmiento-Cano, M. Suárez-Duran, H. Asorey, and L.A. Núñez. [Muon tomography sites for colombian volcanoes](#). *Annals of Geophysics*, **63**(6):661, 2020.

⇒ J. Peña-Rodríguez, J. Pisco-Guabave, D. Sierra-Porta, M. Suárez-Durán, M. Arenas-Flórez, L.M. Pérez-Archila, J.D. Sanabria-Gómez, H. Asorey, and L.A. Núñez. [Design and construction of MuTe: a hybrid muon telescope to study colombian volcanoes](#). *Journal of Instrumentation*, **15**(09):P09006–P09006, (2020).

⇒ A. Vásquez-Ramírez, M. Suárez-Durán, A. Jaimes-Motta, R. Calderón-Ardila, J. Peña-Rodríguez, J. Sánchez-Villafrades, J.D. Sanabria-Gómez, H. Asorey, and L.A. Núñez. [Simulated response of mute, a hybrid muon telescope](#). *Journal of Instrumentation*, **15**(08):P08004, (2020).

⇒ J. Ospino and L. A. Núñez. [Karmarkar scalar condition](#). *Eur. Phys. J. C*, **80**, 166, (2020).

2019

⇒ A. Garcia-Forero, D.A. Villamizar-Mantilla, L.A. Núñez, R.E. Ocazonez, E.E. Stashenko, and J.L. Fuentes. [Photoprotective and antigenotoxic effects of the flavonoids apigenin, naringenin and pinocembrin](#). *Photochemistry and Photobiology*, **95**(4):1010–1018, (2019).

2018

- ⇒ H. Asorey, L.A. Núñez, and C. Sarmiento-Cano. [Exposición temprana de nativos digitales en ambientes, metodologías y técnicas de investigación en la Univ. .](#) *Revista Brasileira de Ensino de Física*, **40**(4), (2018).
- J. Peña-Rodríguez, J. Salamanca-Coy, J.D. Sanabria-Gómez, J. Sánchez-Villafrades, and D. Sierra-Porta. [minimute: A muon telescope prototype for studying volcanic structures with cosmic ray flux.](#) *Scientia et technica*, **23**(3):386–390, (2018).
- ⇒ H. Asorey, S. Hernández-Baraja, F. León-Carreño, L.A. Núñez, J. Peña-Rodríguez, J. Pisco-Guabave, D. Sierra-Porta, and M. Suárez-Durán. [Hardware-level calibration of the chitaga water cherenkov detector in the guane array for space weather study.](#) *Scientia et technica*, **23**(4):563, (2018).
- ⇒ H. Asorey, R. Calderón-Ardila, C.R. Carvajal-Bohorquez, S. Hernández-Barajas, L. Martínez-Ramírez, A. Jaimes-Motta, F. León-Carreño, J. Peña-Rodríguez, J. Pisco-Guavabe, J.D. Sanabria-Gómez, M. Suárez-Durán, A. Vásquez-Ramírez, K. Forero-Gutiérrez, J. Salamanca-Coy, L.A. Núñez, and D. Sierra-Porta. [Astroparticle projects at the eastern colombia region: facilities and instrumentation.](#) *Scientia et technica*, **23**(3):391–396, (2018).
- ⇒ H. Asorey, L. A. Núñez, and M. Suárez-Durán. [Preliminary results from the latin american giant observatory space weather simulation chain.](#) *Space Weather*, **16**(5):461–475, (2018).
- ⇒ M. Ibañez and L.A. Núñez. [On the vortex waves in nonadiabatic flows.](#) *The Astrophysical Journal*, **855**(1):19, (2018).
- ⇒ H. Hernández, L.A. Núñez, and A. Vásquez-Ramírez. Convection and cracking stability of spheres in general relativity. *Eur. Phys. J. C*, **78**(11):883, (2018).
- ⇒ J. Ospino, J.L. Hernández-Pastora, H. Hernández, and L.A. Núñez. [Are there any models with homogeneous energy density?](#) *General Relativity and Gravitation*, **50**(11):146, (2018).

▪ Supervisor of PhD Thesis

2024 ⇒ A. C. Vásquez-Ramírez *Correlation Between Multiple Elves and Storm Dynamics at the Pierre Auger Observatory* **PhD thesis, School of Physics**, Univ. Industrial de Santander, Bucaramanga, Colombia. (Collaborating with R. Mussa)

2021 ⇒ J. Peña-Rodríguez. *Diseño y calibración de un telescopio de muones híbrido para estudios vulcanológicos.* **PhD thesis, School of Physics**, Univ. Industrial de Santander, Bucaramanga, Colombia. (Collaborating with H. Asorey)

2019 ⇒ M. Suárez-Durán. *Variaciones del flujo de radiación cósmica en distintos escenarios geofísicos.* **PhD thesis, School of Physics**, Univ. Industrial de Santander, Bucaramanga, Colombia. (Collaborating with H. Asorey)

2016 ⇒ R.N. Mayol-Arnan. *Modelo para la recolección, normalización y transferencia segura de bitácoras en e-entornos.* **PhD thesis, School of Electric, Electronic and Telecommunication Engineering**, Univ. Industrial de Santander, Bucaramanga, Colombia.

2009 ⇒ Héctor Hernández Guerra, *Ecuaciones de Estado no Locales en Relatividad General.* **PhD thesis, Department of Physics**, Univ. de Los Andes, Mérida Venezuela.

▪ Supervisor of Master's Thesis

2025

⇒ Steven Fernando Rico Aparicio, *Análisis de redes de conectividad cerebral en pacientes epilépticos a través de su representación en espacios de baja dimensionalidad* **Master's Thesis in Physics**, School of Physics, Univ. Industrial de Santander, Bucaramanga, Colombia. (Collaborating with Mario Chávez)

⇒ Eliana Martha Bonalde Marcano *Flood mapping using contrastive learning and semantic segmentation* **Master's Thesis in Applied Mathematics**, School of Physics, Univ. Industrial de Santander, Bucaramanga, Colombia.

2024

⇒ J. Grisales Casadiegos. *Estudio de los efectos de la actividad solar a largo plazo sobre el flujo de rayos cósmicos secundarios en el Observatorio Pierre Auger* **Master's Thesis in Physics**, School of Physics, Univ. Industrial de Santander, Bucaramanga, Colombia. (Collaborating with C. Sarmiento-Cano)

2023

⇒ Y. Domínguez Ballesteros. *Cosmic Neutron Applications for smart agriculture* **Master's Thesis in Physics** University of Rwanda, Kigaly, Rwanda. (Collaborating with C. Sarmiento-Cano)

⇒ A. Martínez-Méndez, *Modelo de Gestión de Recursos Computacionales para Asistir la Reproducibilidad de Experimentos Científicos* **Master's Thesis in System Engineering**, Dept. System Engineering, Univ. Industrial de Santander, Bucaramanga, Colombia.

⇒ D. L. Ramos Salamanca. *Study of magnetic skyrmions in square lattice antiferromagnets* **Master's Thesis in Physics**, School of Physics, Univ. Industrial de Santander, Bucaramanga, Colombia. (Collaborating with Pierre Pujol)

2022

⇒ E. Padilla Expósito *Orbitas de las estrellas S2 y G2* **Master's Thesis in Mathematical Modeling**, Dept. Applied Mathematics, Univ. de Salamanca, Salamanca, Spain. (Collaborating with J. Ospino)

⇒ P. Gutiérrez Benito *Escalares de Estructura en perturbaciones del campo gravitatorio de un agujero negro estacionario* **Master's Thesis in Mathematical Modeling**, Dept. Applied Mathematics, Univ. de Salamanca, Salamanca, Spain. (Collaborating with J. Ospino)

⇒ L.F. Castañeda-Godoy *Propagación Lenta de Discontinuidades en Ambientes de Hidrodinámica y Radiación en Esferas Relativistas* **Master's Thesis in Physics**, School of Physics, Univ. Industrial de Santander, Bucaramanga, Colombia.

2021

⇒ D.F. Suárez-Urango. *Estudio de las condiciones de aceptabilidad física en esferas politropas anisótropas relativistas.* **Master's Thesis in Physics**, School of Physics, Univ. Industrial de Santander, Bucaramanga, Colombia. (Collaborating with J. Ospino)

⇒ P. Serrano-Martínez. *Parametrización de órbitas de partículas mediante escalares de estructura*. **Master's Thesis in Mathematical Modeling**, Dept. Applied Mathematics, Univ. de Salamanca, Salamanca, Spain. (Collaborating with J. Ospino)

2020

⇒ D. A. Rueda Plata. *Modelos de integración de información multimodal en CNN para el análisis de imagen satelital en entornos urbanos* **Master's Thesis in Applied Mathematics**, School of Physics, Univ. Industrial de Santander, Bucaramanga, Colombia. (Collaborating with R. Ramos-Pollan)

2019

⇒ R. Calderón-Ardila. *Estudio de centelladores plásticos en el proyecto mute para muografía de volcanes*. **Master's Thesis in Geophysics**, School of Physics, Univ. Industrial de Santander, Bucaramanga, Colombia. (Collaborating with H. Asorey).

⇒ A. Vásquez-Ramírez. *Estimación de la respuesta generada por el detector MuTe al paso de partículas cargadas*. **Master's Thesis in Physics**, School of Physics, Univ. Industrial de Santander, Bucaramanga, Colombia. (Collaborating with H. Asorey).

2018

⇒ M.A. Vesga-Ramírez. *Inversión geofísica a partir de datos de muografía volcánica para proyecto mute*. **Master's Thesis in Geophysics**, School of Physics, Univ. Industrial de Santander, Bucaramanga, Colombia. (Collaborating with J.D. Sanabria-Gómez)

2016

⇒ L.A. Torres-Niño. *Protocolo de transferencia masiva de datos desde dispositivos hardware hacia repositorios de datos*. **Master's Thesis, School of System Engineering**, Univ. Industrial de Santander, Bucaramanga, Colombia.

2015

⇒ A. Navarro-Noguera *Utilización del concepto de factura para el análisis de la estabilidad de configuraciones esféricas de materia neutra o cargada en Relatividad General*. **Master's Thesis in Physics**, School of Physics, Univ. Industrial de Santander, Bucaramanga, Colombia. (Collaborating with G. González)

⇒ Y. Pérez *Aplicaciones en Meteorología Espacial de los Datos del Proyecto LAGO*. **Master's Thesis in Physics**, Department of Physics, Univ. de Los Andes, Mérida, Venezuela. (Collaborating with H. Asorey).

⇒ M. Suárez-Durán *Modulación de Rayos Cósmicos a Nivel del Suelo por Cambios en el Campo Geomagnético, para la Colaboración LAGO*. **Master's Thesis in Physics**, School of Physics, Univ. Industrial de Santander, Bucaramanga, Colombia. (Collaborating with H. Asorey)

⇒ C. Sarmiento-Cano *Búsqueda de fuentes de astropartículas en los datos de la colaboración LAGO*. **Master's Thesis in Physics**, School of Physics, Univ. Industrial de Santander, Bucaramanga, Colombia. (Collaborating with F. Quiñonez)

1997

⇒ H. Hernández *Movimientos Conformes y Colapso Gravitacional*. **Master's Thesis in Physics**, Department of Physics, Univ. de Los Andes, Mérida, Venezuela. (Collaborating with U. Percoco)

1996

⇒ Félix Aguirre *Ondas de Choque y Esferas Anisótropas en Relatividad General*. **Master's Thesis in Physics**, Department of Physics, Univ. de Los Andes, Mérida, Venezuela.

⇒ Gerardo Contreras. *Colineaciones de Ricci para Métricas con Simetría Esférica*. **Master's Thesis in Physics**, Department of Physics, Univ. de Los Andes, Mérida, Venezuela. (Collaborating with U. Percoco)

1994

⇒ Alejandra Melfo *Esferas Rotantes con Radiación en Relatividad General*. **Master's Thesis in Physics**, Department of Physics, Univ. de Los Andes, Mérida, Venezuela. (Collaborating with L. Herrera)